

1 Newton-Girard Method for Computing the Interacting Kernel

Algorithm 1 Newton-Girard method for computing the interacting kernel

Require: input dimension D
Require: maximum interaction order $\tilde{D}$
Require: base kernels $k_d(\cdot, \cdot)$, $d = 1 \dots D$
Require: order variances σ_ℓ , $\ell = 0 \dots \tilde{D}$
Data: input data X
Ensure: kernel matrix K

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1: for  $d = 1 \dots D$  do
2:    $K_d[i, j] = k_d(x_{i,d}, x_{j,d})$ 
3: end for
4: for  $\ell = 0 \dots \tilde{D}$  do
5:    $S_\ell = \sum_{d=1}^D K_d^\ell$ 
6: end for
7:  $E_0 = 1_{[N,N]}$ 
8: for  $\ell = 1 \dots \tilde{D}$  do
9:    $E_\ell = \frac{1}{\ell} \sum_{k=1}^{\ell} (-1)^{k-1} E_{\ell-k} \odot S_k$ 
10: end for
11:  $K = \sum_{\ell=0}^{\tilde{D}} \sigma_\ell \times E_\ell$ 

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2 Orthogonality in Higher Dimension

For higher order terms, recall OAK uses the product of constrained kernels (equation (9)):

$$\tilde{k}_u(x, x') = \prod_{i \in u} \tilde{k}_i(x_i, x'_i). \quad (1)$$

We show the product of constrained kernel satisfies the orthogonality constraint in higher dimensions, i.e., $\forall i \in u$,

$$\int_{X_i} f_u(x_u) p_i(x_i) dx_i = 0 \quad (2)$$

where each functional component f_u has kernel k_u .

proof. By construction, for each function i with constrained kernel $\tilde{k}_i$, f_i satisfies the orthogonality constraint $S_i := \int_{X_i} f_i(x_i) p_i(x_i) dx_i = 0$ (equation (6)), which implies that:

$$\mathbb{E}_{f_i}[S_i] = 0, \quad \mathbb{V}_{f_i}[S_i] = \int_{X_i} \tilde{k}_i(x_i, x_i) p_i(x_i) dx_i = 0. \quad (3)$$

To prove $\int_{X_i} f_u(x_u)p_i(x_i)dx_i = 0$, it is sufficient to prove the mean and variance of $\int_{X_i} f_u(x_u)p_i(x_i)dx_i$ with respect to f_u is zero. Since we assume f_u has zero mean, the mean $\mathbb{E}_{f_u} \left[\int_{X_i} f_u(x_u)p_i(x_i)dx_i \right] = 0$. The variance

$$\mathbb{V}_{f_u} \left[\int_{X_i} f_u(x_u)p_i(x_i)dx_i \right] = \int_{X_i} \mathbb{E}_{f_u} [f_u(x_u)^2]p_i(x_i)dx_i \quad (4)$$

$$= \int_{X_i} k_u(x_u, x_u)p_i(x_i)dx_i \quad (5)$$

$$= \prod_{j \neq i} k_j(x_j, x_j) \int_{X_i} k_i(x_i, x_i)p_i(x_i)dx_i = 0. \quad (6)$$